

Atlas Reports: agent-drafted client reporting

Every CS manager loses 4+ hours a week stitching seven data sources into spreadsheets and decks. Atlas Reports turns that into a draft waiting in their inbox: the analyst keeps judgment and approval, and gives up the assembly.

STATUS Draft v0.9 — for review	AUTHOR Manthan	DATE 2 Aug 2026	TARGET Beta wk 6 · GA wk 14
REVIEWERS CS Lead · Eng · Design · Data	SURFACE Pepper Atlas, in-product	SOURCES 7 (GSC, GA4, Semrush, GEO, 3× CMS)	COMPANION Sheets 03-04 — Evals

01

PROBLEM

Costs here are modelled, not measured. Sheet 03 replaces them with an instrumented baseline before any target is committed.

The report is cheap. Assembling it is not.

A recurring client report is the same nine moves every time: pull GSC, GA4, Semrush, GEO, check what shipped in the CMS, paste into a sheet, compute deltas, write the narrative, format the deck. Eight are mechanical. The ninth — deciding what mattered and what the client should do — is the only part a client pays for, and it is what gets squeezed when the first eight run long.

- **Capacity.** At an assumed 12 CSMs × 4h, reporting burns ~48 h/week — about 1.2 FTE of margin, and a hard ceiling on accounts per CSM.
- **Consistency.** Every CSM's spreadsheet is a private fork; two clients on one plan get different metrics and different definitions of "traffic".
- **Latency.** Reports arrive on a calendar, not on an event — a ranking collapse surfaces at month-end, weeks after it was actionable.

Atlas already holds the connections and the data. The gap is that nothing turns access into a client-ready artifact without a human doing data entry.

02

USERS

The CS Manager is the only user v1 optimises for. The other two are constraints on the design.

WHO	JOB TO BE DONE	WHAT "GOOD" FEELS LIKE
CS Manager PRIMARY	"Send a report that shows we're earning the retainer — without losing a morning to it."	Opens a draft that is already right; spends the saved time on the two paragraphs of judgment.
CS Lead SECONDARY	"Know every account got a report, on time, at a standard I'd defend."	One view of cadence coverage and draft quality — no chasing.
Client RECIPIENT	"Tell me what changed, why, and what you're doing next."	A short narrative with numbers they can click into; follow-ups answered same-day.

03

GOALS

Non-goals are commitments — each is a place we chose not to spend v1.

GOALS	NON - GOALS (V1)
• Cut analyst time per report by ≥ 50%; ambition 80%. • Make every number in a client report traceable to a source query. • One metric definition per concept, across all clients. • Turn CSM edits into a training signal, not lost work.	• No auto-send. A human approves every report a client sees. • No client-facing chat — Q&A is CSM-facing first. • No deck export: web report, emailed summary, PDF. • No forecasting or budget recommendations. • Auth, billing, dashboard chrome — assumed to exist.

04

SOLUTION SHAPE

Stages 1–3 are deterministic code. Only stage 4 is a model, and it may speak only about findings stage 3 produced.

A pipeline with a model in exactly one place

1 · COLLECT	2 · NORMALISE	3 · DETECT	4 · NARRATE	5 · REVIEW
Pinned pull from bound sources on an as-of date, per-source coverage recorded.	One fact table — entity × metric × period × source. Everything downstream reads only this.	Rules, not vibes. Deltas, rank moves, AI-citation gain/loss, publish→impact, each with effect size.	Grounded LLM writes from findings only. Every sentence carries the metric IDs it used.	Human gate. CSM edits in Report Studio, approves, sends. Edits are logged.

The split matters more than any single stage. Arithmetic done by a language model is unverifiable and occasionally wrong in ways nobody sees until a client catches it; arithmetic done in stages 2–3 is unit-testable and cheap. The model gets the narrow job it is genuinely good at — choosing which of forty true findings are the four worth a client's attention, and saying them in the account's voice.

05

DATA

Black-box endpoints per the brief. Breadth is a cost — each source adds a section a CSM must review. Out of v1: backlink graphs, technical crawl, site audit; they answer engineering questions, not the retainer question.

SOURCE	WHAT WE TAKE	WHY IT EARNS ITS PLACE
GSC	Query + page clicks, impressions, CTR, avg. position	The performance spine; the only source with query-level intent.
GA4	Sessions, engaged sessions, conversions by landing page and channel	Turns visibility into outcomes the client's CFO recognises.
Semrush	Competitor ranks, share of voice, keyword difficulty	Answers "is this us, or the category?" — the question every delta invites.
GEO	AI answer presence, citation share, prompt coverage	Why clients buy Atlas over a rank tracker. Highest-signal section of the report.
CMS ×3	Publish/update events, URL, content ID WORDPRESS · WEBFLOW · CONTENTFUL	Turns "traffic rose" into "traffic rose after we shipped these four pages" — the thread CSMs write by hand today.

06

UX FLOW

Times are relative to the client's scheduled send. The 72-hour lead converts "write a report" into "review a draft". Report Studio is the load-bearing screen: if a CSM cannot verify a claim in one click, they will re-derive it in a spreadsheet, and the four hours come straight back.

Who sees what, when

WHEN	WHO	WHAT HAPPENS
T-72h	System	Pipeline runs on the pinned as-of date; a source below its coverage threshold is flagged, never silently dropped.
T-72h	CS Manager	Notification: <i>"Acme draft ready — 3 findings need your call."</i> Not "a report is ready" — the ambiguous items are named up front.
-72 → -24h	CS Manager	Report Studio. Canvas left, evidence panel right; clicking any number reveals its source, query, period and as-of date. Inline edit, regenerate a section, add a note — every edit captured with a reason tag.
T-24h	CS Manager	Approve → schedules send. Or edit-and-approve, or reject with a reason — which routes into the eval set on Sheet 03.
T-0	Client	Email summary — three findings, one recommendation — linking to the hosted report and a PDF.
T+	CS Manager	Client asks a follow-up. Ask-the-Report answers from the same fact table; the CSM sends, edits, or ignores it.
Weekly	CS Lead	Coverage console: cadence adherence, acceptance rate, most-edited sections.

07

REQUIREMENTS

P0 beta blocker P1
fast-follow P2 later

ID	REQUIREMENT	DETAIL	PRI
FR-1	Report Profile	Per client: brand, competitors (≤5), KPIs, cadence, template, tone, recipients, goals. Set once, versioned.	P0
FR-2	Source binding & health	Bind sources per client; surface last-sync and coverage %. Degradation is stated in the report, never imputed.	P0
FR-3	Metric layer	Single fact model with as-of pinning; re-running a past period reproduces identical numbers.	P0
FR-4	Finding detection	Rules emit candidates with effect size, direction, period and confidence — ranked, not filtered, before the model sees them.	P0
FR-5	Grounded narrative	LLM writes only from candidate findings; each sentence carries metric IDs. An ungroundable sentence is not emitted.	P0
FR-6	Report Studio	Evidence panel, inline edit, section regenerate, manual note, version history.	P0
FR-7	Approval & delivery	Approve → schedule → email, hosted link, PDF. No path to a client that skips approval.	P0
FR-8	Edit capture	Logged as <i>(before, after, section, reason_tag)</i> — both the eval dataset and the improvement loop.	P0
FR-9	Publish → impact join	Attribute movements to CMS content shipped in-window; phrasing capped at "coincided with" unless a pre-registered rule fires.	P1
FR-10	Templates	Monthly Performance, AI Visibility, QBR — with client-level overrides.	P1
FR-11	Ask-the-Report	Grounded Q&A over the same fact table, CSM-facing, under the FR-5 contract.	P1
FR-12	Lead console	Cadence coverage, acceptance rate, overdue drafts, top edit reasons.	P2

08

GROUNDING

A spec, not a guideline — Sheet 03 tests each clause directly.

NON-NEGOTIABLE – THE FIVE CLAUSES

1. No number reaches prose unless it resolves to a metric ID in the fact table. 2. Missing data is disclosed, never imputed — *"Semrush ranks unavailable 12–14 Jul; excluded from the weekly average."* 3. Every report prints its as-of date, and a re-run of that period is identical. 4. Causal language is gated; the default phrasing is correlational. 5. A human approves before a client sees it.

Clause 3 exists because GSC restates recent days for roughly 48–72 hours: without pinned as-of semantics, the same "last month" report yields different numbers on Tuesday than on Thursday.

09

GAPS

Per the brief: what I'd add to the available surface, and the specific thing each unblocks.

WHAT I'D ADD	WHY	UNBLOCKS
Content ID ↔ URL map	The CMSes key on content IDs; GSC and GA4 key on URLs. Without a durable join, publish→impact is string-matching on slugs.	FR-9
Client goal object	"Traffic +12%" is data; "+12%, four points ahead of the Q4 commitment" is the sentence that renews a retainer.	Narrative quality
As-of / snapshot reads	Reproducibility and offline replay both need "the data as it stood on date D", not "the data now".	FR-3, eval L2
Conversion value	Sessions are a proxy; revenue or lead value makes the report legible to the client's finance team.	QBR template

10

ROLLOUT

Sequenced so the measurement instrumentation lands before the thing it measures.

WEEKS	MILESTONE	EXIT CRITERION
0–2	Baseline instrumentation	Two weeks of measured CSM time-per-report. No build starts until it exists.
1–4	Metric layer + detection	FR-2/3/4 pass eval level L0; the 60-report golden set is frozen.
4–6	Narrative + Report Studio	L1 fidelity ≥ 99.5%; shadow reports running on 12 live accounts.
6–14	Stepped-wedge beta	The live experiment reaches its pre-registered stopping point (\$16).
14+	General availability	Ship gate (\$16) met. FR-9/10/11 follow.

11

THE QUESTION

Two questions, not one. Time saved alone is gameable: a tool that emits a plausible report in nine seconds saves four hours and destroys client trust in a quarter.

Did it save the four hours — without costing us the report?

HYPOTHESIS

H1. Atlas Reports cuts median CSM time per approved report by ≥ 50% versus the measured baseline. **H2.** It does so with no degradation in report accuracy, client engagement, or CSAT.

Baseline first. "4+ hours a week" is the stated problem, not a measurement. Weeks 0–2 run an instrumented baseline — in-app timers plus a one-minute weekly diary — establishing minutes-per-report, its variance, and its split across the nine steps. Every target below is expressed against that number.

12

EVAL LADDER

Each level is cheap enough to run at its stated cadence. An eval that runs only before launch is documentation.

LVL	WHAT IT TESTS	METHOD	CADENCE
L0	Metric & detection math	Fixture unit tests per endpoint: deltas, WoW/MoM, position change, share of voice, citation gain/loss. Plus adversarial fixtures — missing source, zero denominator, restated GSC window.	Every commit; blocks merge
L1	Narrative grounding	200 generated claims across 20 report contexts; a checker parses each sentence's metric IDs and recomputes the value. Two failure classes counted separately: <i>ungrounded</i> and <i>mismatched</i> .	Nightly + pre-release
L2	Whole-report quality	60 golden reports, blind pairwise against the human-built report for the same client-period, order randomised. Scored 1–5 on accuracy, completeness, insight, actionability, tone-fit.	Pre-release + monthly
L3	Whether it saves the time	Stepped-wedge field experiment across the CS team (\$15).	Once, 8 weeks

On the L2 judge. Two CS raters and an LLM judge score all 60. The judge is trusted for monthly regression *only* where it reaches Spearman $\rho \geq 0.7$ against them — realistically accuracy and completeness, not insight. Human scoring stays mandatory for the release gate.

13

DATASET

The golden set is the durable asset: a week of CS time that outlives every model we swap in. Its ceiling is measured in \$19.

FACET	THE 60-REPORT GOLDEN SET
Composition	12 clients × 5 historical periods, stratified for what breaks things: 2 enterprise + 2 SMB, 2 with a partial source set, 2 mid-period CMS replatforms, 1 with flat metrics (the hardest narrative to write), 1 with a real incident.
Construction	Freeze a source snapshot as of each period's send date, pair it with the report the CSM sent, and have the owning CSM annotate the 3–5 findings that mattered <i>plus anything the original missed</i> . That last part lets L2 score completeness, not similarity — matching a mediocre report is not the bar.
Growth	Every rejected draft and high-severity edit from FR-8 becomes a case: +10/month, with a quarterly review retiring those the system now passes trivially.

14

METRICS

One north star, decomposed in §17; drivers explain it; guardrails can veto it. Edit distance is a proxy — FR-8 reason tags separate *wrong* from *not how I'd say it*, and only the first counts.

TIER	METRIC	INSTRUMENT	TARGET
North star	CSM minutes per approved report	In-app session timer + weekly diary	≥ 50% below baseline; stretch 80%
Driver	Draft acceptance rate	Approved with < 10% token edit distance	≥ 70% by week 8
Second-order	Capacity headroom	Modelled from time freed (\$18)	Tracked for real post-GA
Quality gate	Numeric fidelity	L1 automated citation checker	100% grounded · ≥ 99.5% exact
Quality gate	Unsupported claim rate	L2 human audit, full golden set	< 1 per 20 reports
Guardrail	Client-flagged errors	CS ticket tag <code>report_error</code>	No increase vs. baseline
Guardrail	Report open & click rate	Email telemetry	No drop > 5 pp
Guardrail	Client CSAT, single item	Post-report survey, quarterly	No decrease

15

EXPERIMENT

Randomising by client would leak: a CSM who gets faster on one account gets faster on all of them. Contamination — control CSMs hearing about the tool — is tracked weekly and treated as bias toward the null.

Stepped-wedge, clustered on the CS manager

- Design** — 8 weeks, 4 cohorts of CSMs crossing over every 2 weeks in randomised order. Everyone ends up treated: that protects adoption, gives a within-subject contrast, and separates the effect from seasonality and the month-end peak.
- Outcome** — minutes per approved report, via a pre-registered mixed-effects model `minutes ~ treated + week + (1|csm)`, $\alpha = 0.05$ two-sided, reported as an effect with its 95% CI rather than a bare p .
- Power** — at between-CSM SD ≈ 25 min and ICC ≈ 0.25, ~14 CSMs × 8 weekly observations gives > 80% power for a 25-minute reduction. **Under 10 CSMs the wedge is underpowered** — fall back to a within-subject interrupted time series, ≥ 6 pre and ≥ 6 post weeks each.

16

DECISION

Written before the data arrives, so the result is read rather than negotiated. Stated formally in §20.

VERDICT	PRE-REGISTERED CONDITION
Ship	Time reduction ≥ 50% with CI lower bound > 25%; every quality gate met; no guardrail breach; L2 overall ≥ human baseline – 0.3, accuracy ≥ 4.5/5.
Hold	Any quality gate missed, whatever the time saving. A wrong number in front of a client is the one failure that does not average out.
Rethink	By week 8, acceptance < 40% and edit distance not trending down — the insight layer has not earned trust. Fall back to <i>assembly-only</i> : the pipeline builds the data sections, the CSM writes the narrative. That still removes eight of the nine steps.

KNOWN LIMIT — eight weeks measures time and quality, never retention or account growth. Those need a year and more accounts than we have, and are tracked as observational lagging indicators, not evidence of causation.

17

TIME MODEL

Sheet 03 asks whether the four hours fell. This asks which hour fell — the only version of the question that can distinguish success from a CSM rubber-stamping.

Where the four hours actually go

A single minutes-per-report number cannot tell *we removed the assembly from the CSM stopped reading*, and the guardrails that would catch the second — CSAT, client-flagged errors — are quarterly and rare. So the north star is banded across the nine moves from §01, and each band carries its own target.

BAND	MOVES IT COVERS	TARGET	INSTRUMENT
Collection	Pull GSC, GA4, Semrush, GEO; check the CMS	→ 0	Pipeline replaces it entirely
Assembly	Paste into the sheet, compute deltas	→ 0	Metric layer (FR-3) replaces it
Composition	Write the narrative, format the deck	Format → 0; narrative → review-and-edit	Studio canvas time
Judgment	Decide what mattered; decide what to recommend	Held flat or up	Evidence-panel + edit time

Instrumentation. The in-app timer already attributes session time by active surface — source panel, studio canvas, evidence panel — which maps onto the bands without asking the CSM for anything extra. The weekly diary supplies one corroborating split. Weeks 0–2 measure the real shape; until then the bands have targets but no baseline.

THE SHAPE TEST

If total time falls but the judgment band falls with it, that is a **failure, not a success** — we will have automated the thinking and kept the typing. A report that takes 18 minutes with 12 of them spent on judgment is the win. Eighteen minutes with two is the failure mode this metric exists to catch.

18

CAPACITY

The metric the business buys. It is second-order here because it physically cannot move inside an eight-week trial.

FACET	ACCOUNTS PER CSM AT CADENCE-COMPLIANT QUALITY
In the trial	Modelled, not measured. Nobody hands a CSM new accounts mid-wedge, so capacity headroom is inferred: time freed per CSM-week ÷ time per account-week. The wedge measures time; capacity is the inference drawn from it, and the document should say so rather than dress an inference as a result.
Post-GA	Becomes a real tracked metric — accounts per CSM and reports per CSM-week, quarterly, with the expected trajectory pre-registered before GA so it cannot be re-fitted afterwards.
Failure mode	Capacity bought with cadence slippage or quality drop. The §14 quality gates travel with this metric, and FR-12 cadence adherence is the check: headroom claimed while adherence falls is not headroom.
Operational SLO	Time to first draft ≤ 4 min at p90 (pipeline telemetry). Not a success metric — a latency floor, since a draft that misses the T–72h window costs a CSM the whole review lead.

19

HUMAN CEILING

§13 compares against the report the CSM sent — a floor. This measures the ceiling, and the distance between two competent humans.

What "human-level" actually scores

STEP	PROTOCOL
Sample	15 of the 60 golden cases from §13, stratified across the same archetypes.
Procedure	Three senior CSMs write each case independently from the same frozen snapshot, blind to one another and to the report originally sent.
Cost	~90 min per report per rater ≈ 67 senior-hours , run once, refreshed annually or on a major model change. Stated plainly because a protocol whose cost is hidden is a protocol that never gets run.
Outputs	Human mean and SD per rubric dimension — the ceiling <i>and</i> the noise floor. Inter-rater agreement (ICC or Krippendorff's α). Finding overlap: how often two humans independently pick the same 3–5 findings.
Use	The target stops being "beat the historical report" and becomes " land inside the human band ". Where two competent humans disagree, an agent that differs from one of them has not necessarily erred.

It also disciplines §12: a rubric dimension where three senior CSMs cannot agree is a dimension no LLM judge should be trusted on either, and probably not one we should gate a release on. Expect finding overlap to come in lower than intuition suggests — which reframes a good share of "the agent missed something" as ordinary human variance.

20

DECISION RULE

§16 gives the verdicts. This gives the test behind them, so the margin is derived rather than chosen after the fact.

ENDPOINT	PRE-REGISTERED TEST
Quality NON-INFERIORITY	One-sided. $H_0: \mu_{\text{agent}} \leq \mu_{\text{human}} - \delta$ against $H_1: \mu_{\text{agent}} > \mu_{\text{human}} - \delta$, with $\delta = 0.5 \times \text{the inter-human SD measured in §19}$ — half the noise between two competent humans, rather than the round 0.3 §16 quotes. Accuracy carries a tighter δ and an absolute floor: it is the one dimension where sitting inside the human band is not sufficient.
Time SUPERIORITY	One-sided, from the §15 mixed-effects model. Ship requires the CI lower bound above the stated MDE, not merely a point estimate that clears it.
Both	Required together. Non-inferior quality alone is not a win; faster alone is certainly not. Because two primary endpoints must <i>both</i> pass, this is an intersection-union test and takes no alpha adjustment.
Revealed preference	After the wedge, two weeks optional. Ship needs ≥ 70% continued voluntary use . It prices what the rubric cannot — trust, fit, and whether the thing is pleasant to use.

EVERYTHING ELSE IS EXPLORATORY — segment cuts, per-template breakdowns and archetype-level results are labelled as such when reported. Pre-registration is only worth something if the list stays short.